

ULAANGOM, Mongolia

WMO: 442120

Lat: 49.9714N

Lon: 92.0778E

Elev: 3081

StdP: 13.13

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.4	-38.8	-49.2	0.3	-42.2	-45.0	0.4	-38.5	11.8	-24.3	10.4	-26.5	3.3	180	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	86.6	60.9	83.4	60.0	80.6	59.1	63.9	79.5	62.3	77.6	61.0	75.7	7.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.5	82.0	69.4	56.6	76.7	68.7	54.9	71.8	68.2	30.9	79.1	29.6	78.0	28.6	75.7	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	15.1	12.3	-43.5	92.4	4.6	3.8	-46.8	95.1	-49.5	97.3	-52.0	99.5	-55.4	102.2		
(5)			-43.6	66.6	4.5	2.3	-46.8	68.3	-49.5	69.6	-52.0	70.9	-55.3	72.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	-23.9	-18.2	3.8	38.0	53.0	64.7	68.5	63.7	50.8	35.6	16.1	-10.3
	DBStd	33.47	7.01	9.13	12.18	10.75	8.12	6.47	5.12	5.82	7.32	7.54	9.99	11.21
	HDD50	9488	2291	1910	1434	377	63	1	0	0	77	449	1018	1868
	HDD65	13514	2756	2330	1899	811	377	82	23	94	429	912	1468	2333
	CDD50	1719	0	0	0	15	158	443	574	426	101	2	0	0
	CDD65	269	0	0	0	0	7	74	132	54	2	0	0	0
	CDH74	2867	0	0	0	1	135	880	1265	551	35	0	0	0
	CDH80	728	0	0	0	0	20	224	355	126	3	0	0	0
Wind	WSAvg	4.8	2.5	2.9	3.9	5.9	7.1	6.4	5.7	5.6	5.7	5.1	4.3	2.9
Precipitation	PrecAvg	5.8	0.1	0.1	0.2	0.2	0.3	0.9	1.5	1.3	0.4	0.3	0.4	0.2
	PrecMax	9.8	0.2	0.3	0.4	0.6	0.9	2.7	4.8	3.5	1.2	1.2	0.8	0.6
	PrecMin	2.8	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.2	0.0	0.0	0.0	0.0
	PrecStd	2.1	0.1	0.1	0.1	0.2	0.2	0.6	1.3	0.9	0.3	0.3	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.7	11.7	37.0	70.1	83.0	89.8	92.1	88.1	78.6	63.1	40.9	17.9
		MCWB	-3.8	10.2	31.6	48.8	57.6	61.0	64.6	61.3	57.3	46.4	34.4	16.2
	2%	DB	-6.1	4.5	30.7	64.7	77.9	85.6	87.5	83.4	73.1	57.0	36.1	14.6
		MCWB	-7.0	3.3	27.4	46.4	54.8	59.4	61.8	60.9	55.0	44.3	31.4	13.5
	5%	DB	-9.1	0.3	26.7	60.1	73.6	82.3	84.0	79.9	68.6	52.9	32.6	10.2
		MCWB	-9.8	-0.7	24.0	44.3	52.2	58.6	60.7	59.6	53.1	42.4	29.2	8.9
	10%	DB	-12.1	-3.6	22.6	55.6	69.4	79.3	80.9	76.5	65.0	49.1	29.6	6.4
		MCWB	-12.6	-4.4	20.8	42.0	50.2	57.6	60.1	58.8	51.0	40.1	27.1	5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.6	10.0	31.9	50.0	59.1	63.9	67.8	65.1	58.5	48.0	34.9	16.1
		MCDB	-3.0	11.4	37.1	67.6	79.5	83.6	84.2	80.4	75.0	60.5	39.8	17.1
	2%	WB	-6.8	3.5	27.8	47.3	55.9	61.9	64.9	62.9	56.0	45.2	31.8	13.4
		MCDB	-6.1	4.5	30.6	63.4	75.5	79.6	80.3	77.7	71.0	55.6	35.3	14.6
	5%	WB	-9.8	-0.5	24.3	44.8	53.4	60.3	63.1	61.4	53.9	43.0	29.5	8.9
		MCDB	-9.2	0.4	26.5	58.2	71.4	76.7	78.1	75.4	67.4	52.2	32.4	10.0
	10%	WB	-12.7	-4.4	20.7	42.5	51.1	58.9	61.7	60.0	51.8	40.4	27.2	5.4
		MCDB	-12.2	-3.7	22.8	54.6	67.4	75.3	76.4	73.4	63.8	48.6	29.5	6.3
Mean Daily Temperature Range	5% DB	MDBR	16.4	19.9	19.9	19.2	23.3	22.8	20.5	20.9	22.3	20.1	12.9	13.4
		MCDBR	17.8	21.7	15.9	26.8	29.0	27.4	25.4	25.4	26.0	24.8	16.0	11.0
		MCWBR	17.0	20.7	12.9	12.8	12.9	9.9	8.7	9.6	12.4	15.0	12.0	10.0
	5% WB	MCDBR	17.5	21.7	14.8	25.4	27.3	23.1	21.0	21.9	24.8	24.1	15.5	10.6
		MCWBR	17.0	20.8	12.1	12.7	13.0	9.6	9.0	9.5	12.5	15.2	11.9	9.8
Clear-Sky Solar Irradiance	taub	0.250	0.278	0.326	0.373	0.374	0.377	0.375	0.355	0.317	0.300	0.274	0.250	
	taud	2.284	2.241	2.219	2.220	2.262	2.348	2.388	2.412	2.459	2.424	2.310	2.258	
	Ebn at Noon	256	274	278	275	280	279	278	279	280	264	241	236	
	Edh at Noon	21	29	36	41	41	38	36	33	28	24	21	19	
All-Sky Solar Radiation	RadAvg	430	745	1227	1716	2070	2135	2024	1781	1415	890	447	326	
	RadStd	24	43	60	72	107	84	116	90	65	51	37	25	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (6 neighbors)		N/A	N/A	-4.37	N/A	-1.10	-1.33	N/A	N/A	N/A
		N/A	N/A	-4.61	N/A	-1.55	-2.09	N/A	N/A	N/A

Nomenclature: See separate page